DE–4216 **11**

DISTANCE EDUCATION

M.Sc. (ZOOLOGY) DEGREE EXAMINATION, MAY 2012.

ANIMAL DIVERSITY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following in about 300 words each

1. Explain the different organization of tissues and their function in animals.

2. Write an essay on different theories of taxonomy.

3. Give an account on the phylogeny and adaptive radiation of Ctenophora.

4. Write a note on the asexual reproduction of parasitic Protozoa.

5. What are the general characters of insecta and explain with suitable examples.

6. Describe the structural and functional organization of fishes.

7. Write an essay on origin, relationships and adaptation of birds.

8. Give an account on human evolution with phylogenetic history.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR of the following questions in about 500 words each.

9. Classify the major sub-division of animal kingdom and explain.

10. Describe the ecological relationship, form and functions of porifera.

11. Give a detailed account on body plan of Polycheata and Oligocheata.

12. Discuss in detail the origin, evolution and diversity of the bony fishes.

13. Write an essay on the early evolution of terrestrial vertebrates.

14. Summarize the phylogeny of chordate.

15. Describe the origin of flights in birds.

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2012.

CELL AND MOLECULAR BIOLOGY

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

All questions carry equal marks.

1. How can you distinguish between a prokaryotic and eukaryotic cell?

2. Write short notes on unit membrane model of plasma membrane.

3. Differentiate the properties of prokaryotic DNA polymerases.

4. Give an account on synthesis of leading and lagging strands.

5. Explain the role of transcription in prokaryotes.

6. Write short notes on RNA splicing.

7. What are post-translational modifications of proteins?

8. How does protein synthesis differ in prokaryotes and eukaryotes?

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

All questions carry equal marks.

9. Describe the ultra structure of a mitochondrion and compare the properties of the inner and outer membranes.

10. Describe the various stages of the cell cycle with the specific duration.

11. Explain the characters between the three different classes of RNAs.

12. Write in detail the eukaryotic transcriptional promoters, enhancers and silencers.

13. Describe the mechanism of translation during protein synthesis in a prokaryotic cell.

14. Explain various methods of regulation of gene expression in prokaryotes.

15. Compare and contrast the role of inducer and co-repressor in lactose and tryptophan operons in *E.coli*.

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2012.

GENETICS AND EVOLUTION

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions, each in about 300 words.

1. Describe the interaction of genes with complementary and supplementary factors.

2. Explain the phenomenon of crossing over with example.

3. Describe the fine structure of gene.

4. Discuss the inborn errors of metabolism.

5. Explain sequential expression of genes with examples.

6. Discuss the role of polymorphism in evolution.

7. Explain the modes of speciation.

8. Write short notes on

   (a) Molecular drive

   (b) Molecular variation.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR of the following questions, each in about 500 words.

9. Give an account on the structural and numerical changes in chromosomes.

10. Define induced mutation. How can you detect? Add a note on its significance

11. Describe the fine structure of gene with illustrations.

12. Prove the role of natural selection with evidences.

13. What is phylogenetic tree? Discuss its application molecular genetics.

14. Write an essay on the role of genes in development and differentiation with examples.

15. Give an account on Neo-Lamarckism.

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2012.

BIOCHEMISTRY AND ANIMAL PHYSIOLOGY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

All questions carry equal marks.

1. Describe the different structures of Protein.

2. Explain the metabolic importance of Vitamin B group.

3. Write an account on the biologically importance of monosaccharides.

4. Explain the Krebs cycle.

5. Describe the ultrastructure of a skeletal muscle.

6. Explain the mechanism of hearing in mammal.

7. Write an account on temperature regulation in animals.

8. Elucidate the occurrence of biological clock in organisms.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following.

All questions carry equal marks.

9. Explain the mechanisms of hormone action, the membrane receptors and elaborate on the role of cyclic AMP.

10. Describe the synthesis of glucose from non-carbohydrate sources.

11. Explain the essential features of β -oxidation scheme.

12. Describe the transportation of respiratory gases.

13. Explain the mechanism of conduction of nerve impulse.

14. Explain the role of visual pigments in colour vision.

15. Write an essay on Osmoregulation in animals.

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2012.

BIOPHYSICS AND BIOSTATISTICS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

All questions carry equal marks.

1. Explain the types of bonds in biomolecules.

2. Enumerate the properties of electromagnetic radiation.

3. Describe the working principle of LASER.

4. Write an account on the effect of radiations on macromolecules.

5. Explain the working principle of gas-liquid chromatography.

6. Explain the technique of agarose gel electrophoresis.

7. Explain the various types of diagrammatic representation of biological data.

8. Describe the types and application of regression analysis.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

All questions carry equal marks.

9. Explain the technique of X-ray diffraction crystallography.

10. State the principle and application of laws of thermodynamics.

11. Explain the Beer and Lambert's laws of light absorption.

12. Describe the various types of blotting techniques.

13. Explain the instrumentation technique of Gieger-Muller counter.

14. Explain the various methods of collecting primary and secondary data.

15. Calculate mean and standard deviation for the following data :

Number of eggs laid :	1-3	3-5	5-7	7-9	9-11
Number of insects :	15	18	27	10	6

DE–4221 **21**

DISTANCE EDUCATION

M.Sc. DEGREE EXAMINATION, MAY 2012.

Zoology

ENVIRONMENTAL BIOLOGY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions in about 300 words each.

1. Discuss the process of food web in a pond ecosystem with suitable examples.

2. Describe in detail about ecological pyramids.

3. Discuss the ecological significance of recycle pathways.

4. Explain monoclimax and polyclimax theories with suitable examples.

5. What is community? Explain the structure of community.

6. Why are coral reefs so important?

7. How deforestation induce global warming?

8. What protection measures in India are taken regarding environment?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions
in about 500 words each.

9. What do you understand by ecosystem? Describe the various components of an ecosystem giving suitable examples.

10. Give an account of Leibig's law of minimum and Shelfords law of tolerance and the combined concept with an example.

11. Describe different types of biogeochemical cycles. Discuss their significance

12. Describe the factors that play key role in population regulation.

13. What do you understand by renewable and non - renewable sources of energy? Give an account of non - renewable sources of energy.

14. Describe the ecological features and adaptations to freshwater habitat.

15. What is air pollution? What are the different sources of air pollution? Describe the effects of air pollution on human life and plants.

DE–4222 **22**

DISTANCE EDUCATION

M.Sc.(Zoology) DEGREE EXAMINATION, MAY 2012.

DEVELOPMENTAL BIOLOGY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

Each questions carry equal marks.

1. Neatly Sketch and explain the general structure of sperm.

2. How will you classify the eggs? Explain.

3. Enumerate the factors affecting cleavage.

4. Discuss the mechanism of morphogenetic movements.

5. What is Emboly? Explain the various types of Emboly movements.

6. Briefly explain about foetal membrane in chick.

7. What is induction? Explain the mechanism of induction

8. What are the factors Involved in teratogenesis?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

Each questions carry equal marks.

9. Discuss on different stages of spermatogenesis.

10. Write an essay on plane and patterns of cleavage.

11. Discuss on presumptive organ forming areas in frog.

12. Give an account on gastrulation in chick.

13. Describe the type's placenta in mammals.

14. Write a detailed account on concepts of Assisted Reproduction Technologies.

15. Discuss the cellular basis of morphogenesis.

DE–4223 **23**

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2012.

MICROBIOLOGY AND IMMUNOLOGY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions, each in about 300 words.

1. Give an account on microbial diversity.

2. Explain the classification of bacteria based on size and shapes.

3. What are types of microbial culture media? Add a note on their importance in microbiology.

4. List out the microbes of milk and explain the methods to prevent their growth.

5. Explain the functions of IgG.

6. Classify antigens based on any two methods and give suitable examples.

7. What are vaccines? Explain briefly the importance of vaccine schedule.

8. Elucidate the principle and applications of Widal test.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions, each in about 500 words.

9. Describe the ultrastructure of Gram-negative bacteria with neatly labelled diagrams.

10. Explain any three molecular tools employed to identify microbes.

11. Give a detailed account on etiology of any two types of bacterial infection in human.

12. Write an essay on various types of immune cells and their functions.

13. Explain the mechanisms of antigen-antibody interactions.

14. How does immune system react against tumors?

15. Explain the principle and applications of various types of ELISA.

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DE–4224 **24**

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2012.

ANIMAL BIOTECHNOLOGY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

All questions carry equal marks.

1. Give a detailed account on the various enzymes involved in genetic engineering.

2. What are RAPD and RFLP? Write down their applications.

3. Briefly explain the physicochemical parameters of culture media.

4. What is vermiculture? Give an account on indoor and outdoor vermiculture.

5. Write a note on the physiology of silkglands of *Bombyx mori*.

6. Explain the techniques involved in maintaining of animal cell culture.

7. Briefly describe the *in vitro* production and purification of insulin.

8. Give an account on protein sequencing data base.

SECTION B — ($4 \times 15 = 60$ marks)

Answer any FOUR of the following.

All questions carry equal marks.

9. Explain various DNA fingerprinting techniques with their role in biotechnology.

10. Discuss in detail the different methods involved in animal tissue culture.

11. What is vermicompost? Explain the method of preparation of vermicompost.

12. Describe in detail the various species of earthworms and their breeding methods.

13. Write an essay on the cultivation of mulberry and silkworm rearing.

14. What are stem cells? Write a detailed note on the role of embryonic and adult hemopoietic stem cells.

15. Discuss the role of search engines in bioinformatics. Explain few popular search engines.

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DE–4225 **251**

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION, MAY 2012.

FISHERIES AND AQUACULTURE

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

All questions carry equal marks.

1. Freshwater fishes -Classify.

2. Write a short note on

   (a) Dugout canoes and

   (b) Catamarans.

3. Write an account on food and feeding habits of freshwater cultivable fish.

4. How will you select and prepare a pond for fish culture.

5. How do you prepare artificial feed for shrimp? Explain.

6. Discuss the advantages of paddy cum aquaculture.

7. Give an account on common sea weeds and their control.

8. Write a detailed account on the principles and process of fish preservation.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

All questions carry equal marks.

9. Discuss in detail the economics of aquacu1ture.

10. Write a detailed account on fishery by products and their uses.

11. "Water quality management is one of the important aspects of aquaculture" – Discuss.

12. Write an account on mono-sex culture. Add a note on the problems and prospects of mono sex culture.

13. Write a detailed account on symptoms and treatment of protozoan diseases.

14. Induced breeding is a revolutionary technique- Explain.

15. Discuss the examination of freshness of fishes by using biochemical methods.

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DE–4226 **252**

DISTANCE EDUCATION

M.Sc. (Zoology) DEGREE EXAMINATION,
MAY 2012.

PARASITOLOGY

Time : Three hours Maximum : 100 marks

PART A — (5×8 = 40 marks)

Answer any FIVE questions.

All questions carry equal marks.

1. How are ectoparasites differ from endoparasites? Discuss with suitable examples.

2. Write an account on Schistosomiasis.

3. Enlist and explain the characteristic properties of Trematodes.

4. Give a detailed account on Amoebiasis.

5. Explain the pathophysiological mechanism of trypanosome.

6. Discuss the symptoms and medical care for Filariasis.

7. Give an account on Bed Bugs and its vector role.

8. Explain the CDC recommendation for malaria prevention.

PART B — ($4 \times 15 = 60$ marks)

Answer any FOUR questions.

All questions carry equal marks.

9. Explain the Factors Contributed to the Widespread Parasite Epidemic, increasing the risk of more people acquiring parasite infections.

10. Describe the life cycle of Plasmodium.

11. Differentiate and explain the life cycle of Liver fluke and lung fluke with suitable illustration.

12. Explain the molecular based diagnostic approaches in parasitology.

13. Explain the biology and Morphology of Tsetse fly.

14. Discuss the diagnosis, treatment and prevention for Giardia.

15. Discuss the negative role of mites which affects animal health.